

Satyam Kumar

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EXPERIENCE

Tata 1mg

Gurugram, India

Senior Software Development Engineer (SDE III) – AI Systems & Platform

April 2026 – Present

- Authored a persistent multi-agent orchestration framework (extending PydanticAI) with durable session persistence, resumable/branchable threads, and context compaction – adopted as the core orchestration layer across **6+** internal AI products
- Built long-lived autonomous coding-agent infrastructure supporting async execution, real-time progress streaming, and pause/resume/cancellation for complex multi-step coding tasks
- Extended **DeputyDev**'s code-intelligence layer: Tree-sitter-based parsing/chunking, semantic vector search with symbol/function-level retrieval, and multi-provider LLM routing (OpenAI, Anthropic, Google) with fallback
- Built **1mg MCP**, an authenticated MCP server exposing commerce tools (product search, cart, checkout, order tracking) with location-aware context enforcement, and an AI-assisted catalog-management system (40+ entity types) with mandatory human-in-the-loop approval
- Led cross-functional adoption of GenAI tooling across 1mg's engineering org (**30%+** throughput gain); co-authored a published enterprise study on the platform's productivity impact (see Publications)

Tata 1mg

Gurugram, India

Software Development Engineer II

April 2024 – April 2026

- Built and led **DeputyDev**, an AI-powered developer platform (code generation, code review, productivity analytics) from inception to **300+ engineers** deployed org-wide via the VS Code Marketplace
- Overhauled query-solving, history, and intent-resolution systems, improving response speed **3–4x** and cutting chat error rates from **300+ to under 100/day**
- Re-architected the payments/refunds pipeline (**50%** less memory, **30%** lower latency) and led delivery of **Lab Buddy**, a vector-search-based diagnostics assistant; enhanced Care Plan/Loyalty systems, driving **6x** adoption growth

Tata 1mg

Gurugram, India

Software Development Engineer I

October 2022 – April 2024

- Built a self-serve onboarding system for third-party partnerships and enhanced the loyalty/rewards service for fault tolerance, cutting integration implementation time by **30x**

Yellow Class (now SpeakX)

Remote (later Gurugram), India

Software Development Engineer

July 2021 – October 2022

- Built microservices for a top-10 Indian education app, enabling **5x** growth in user interactions; designed the Quiz module handling **1,000+ requests/min** with minimal resource footprint

PUBLICATIONS

Intuition to Evidence: Measuring AI's True Impact on Developer Productivity – Co-author. Year-long, 300-engineer enterprise study on AI-assisted development; found a **31.8%** reduction in PR review cycle time and a **28%** increase in production code shipment volume. *arXiv:2509.19708, 2025.*

OPEN SOURCE

Google Summer of Code – Kiwix | Python, Offliners, Zimfarm

Jun 2020 – Aug 2020

- Selected among 1,199 global participants; improved Python offliners and Zimfarm to archive web content into ZIM files, achieving **10x** faster offlining and up to **50%** smaller output files across seven projects

EDUCATION

Indian Institute of Information Technology, Kalyani

Kalyani, India

B.Tech in Computer Science and Engineering; CGPA: 9.26/10

Expected June 2021

Honors: 2025 Dept. Award for Performance, Tata 1mg (DeputyDev/GenAI) | 2020 National Winner, Smart India Hackathon

TECHNICAL SKILLS

AI / ML: Multi-agent orchestration, agentic workflows, LLM routing & fallback, RAG, semantic & vector search, prompt engineering & caching, multimodal inference, model evaluation

Frameworks: PydanticAI, LangGraph, Model Context Protocol (MCP), LLM observability (Langfuse)

Languages: Python, JavaScript, TypeScript, SQL

Infrastructure & Data: PostgreSQL, MongoDB, Redis, Kafka, RabbitMQ/SQS, Qdrant, Weaviate, API governance